GRAIL

Compact Multimodal Similarities for Astronomy Citation Linking

Egor Fedosov | Ural Federal University

0.48
Local OOF macro-F1
Fixed factorized model

0.48
Kaggle macro-F1
Displayed 19 Sep 2026*

+0.068
Gain over raw cosines
Same boosted classifier

190
Symmetric pair features
Frozen encoders

01 Task & evaluation

Can astronomy figures and captions reveal a **direct citation relation**, beyond shared subject matter?

10,000 training + 10,000 scoring pairs

4,000 figure-caption
3,000 figure-figure | 3,000 caption-caption

Four labels: same figure; same paper; citation-related papers (either direction); unrelated papers. Same figure is feasible only for figure-caption pairs.

Validation separates exact objects

Five outer folds assign entire connected components of the object-pair graph. **Zero exact-object overlap** between training and validation. TF-IDF, IDF and representation statistics are fitted within each training partition.

Scope of the claim
Exact-object grouping does not ensure source-paper separation. Paper-disjoint evaluation remains future work.

02 Compact multimodal architecture

Unordered pair of objects
Figure-caption / figure-figure / caption-caption

Dual CLIP
ViT-B/16 + ViT-B/32
Images & captions
Frozen, 512-D each

SciNCL
Caption pairs only
Scientific text
Frozen, 768-D

Explicit evidence
OCR + lexical overlap
TF-IDF + rarity
33 image descriptors

190 symmetric pair features
3 modality + 34 lexical + 14 TF-IDF + 5 length
+ 68 representation statistics + 66 layout coordinates

Factorized gradient boosting
Separate model for each pair modality
Same figure? | Same paper? | Citation-linked?

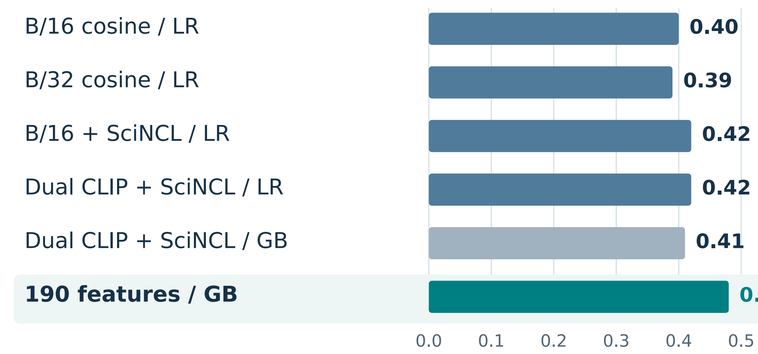
7 leaves per tree | 180 iterations | learning rate 0.055

Conditional probabilities combine into four class probabilities.
No encoder fine-tuning; fixed model uses no decision offsets.

03 Results & controlled comparisons

Overall local OOF macro-F1

Four classes, 10,000 pairs, same outer folds



LR: logistic heads. GB: fixed seven-leaf boosted heads.

Removing evidence lowers macro-F1

Removed feature block	Absolute drop
OCR pipeline	**0.0081**
TF-IDF / rarity	**0.013**
CLIP B/16 block	**0.0096**
CLIP B/32 block	**0.0076**

CLIP removals use the replayed full model; its OOF score differs from the saved fixed run by 0.00022.

KEY FINDING Explicit pair evidence improves a compact frozen-encoder system. **Citation-related papers remain hardest (F1 0.37)**; nested selection gives no observed aggregate gain.

* Kaggle scores are those displayed on 19 September 2026, not final private-leaderboard results. Scores and differences retain the paper's two-significant-digit reporting; differences were computed before rounding.

References: Grezes et al. (2026), AstroCLIMB / WASP; Radford et al. (2021), CLIP / ICML; Ostendorff et al. (2022), SciNCL / EMNLP. Results: GRAIL paper v4, Tables 1-5.

CODE & EXPERIMENTS
github.com/denzasikora-lab/**astroclimb-grail**